

LAB Logbook

Lab 1

```
▶ #SID is 2520731, so the last two digits are 31.

#1)Create a vector using np.arange.
a = np.arange(31)
print(a)

#2)Change a to a 2-D array with 1 row
a1 = a.reshape(1, 31)
print(a1)

#3)Save it in another array. Print the array.
a2 = a1
print(a2)

#4)Check the shape attribute value.
print(a2.shape)
```

... [0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
24 25 26 27 28 29 30]
[[0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
24 25 26 27 28 29 30]]
[[0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
24 25 26 27 28 29 30]]
(1, 31)

Lab 2

```
# 1. Determine n from the last digit of the SID=2520731
n = 1

# 2. Group by relationship and hours-per-week
Group_by_relationship_hours = data.groupby(["relationship", "hours-per-week"])
Group_by_relationship_hours.size()

# 3. Reduce hours-per-week by n using a function
def reduce_hours(x):
    return x - n

data["hours-per-week-reduced"] = data["hours-per-week"].apply(reduce_hours)

data

# 4. Group by relationship and reduced hours-per-week
Group_by_relationship_reduced_hours = data.groupby(
    ["relationship", "hours-per-week-reduced"]
)

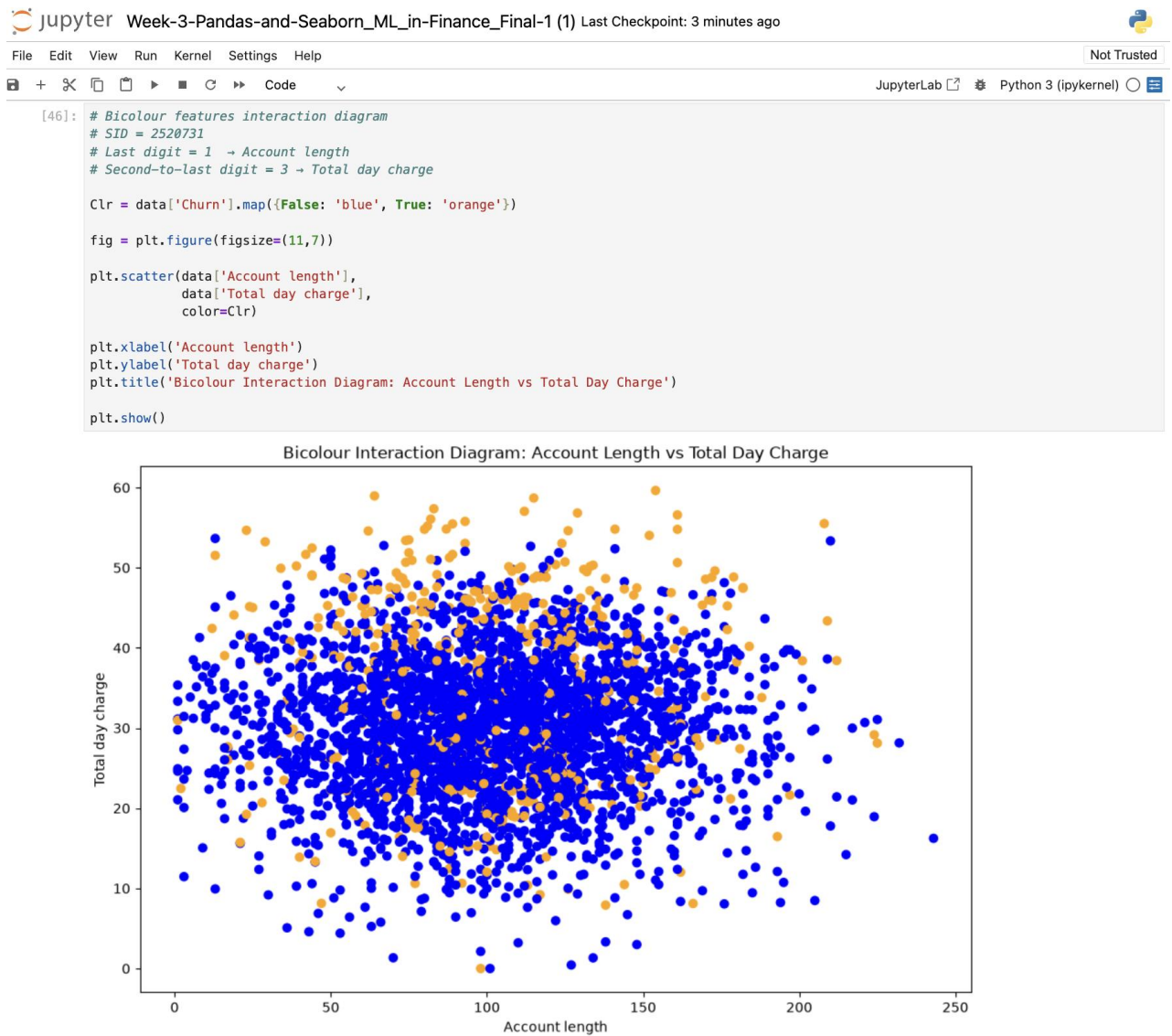
Group_by_relationship_reduced_hours.size()
```

...

		0
relationship hours-per-week-reduced		
Husband	12.0	1
	39.0	2
	44.0	1
	79.0	1
Not-in-family	15.0	1
	39.0	2
	49.0	2
Own-child	29.0	1
Wife	39.0	2

dtype: int64

Lab 3



Lab 4

Lab 5

Lab 6

Lab 7

Lab 8

Lab 9

Lab 10

Lab 11

Lab 12